

CAPSTONE PROJECT PROPOSAL

**Preventing Data Breaches and Record Inaccuracies in Public Libraries Through a
Cybersecurity-Enhanced Management System**

In partial fulfillment for the requirements in Capstone 1 and Research

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Cyber Security Track

Second Semester 2025-2026

CHAPTER 1

INTRODUCTION

1.1 Background Of The Study

Public libraries play an essential role in providing access to information, educational resources, and community services for students and the general public (Singh & Kumar, 2025). As technology continues to evolve, many libraries are transitioning from manual record keeping to digital management systems to improve accessibility and operational efficiency (Swaminathan, 2024). This shift allows libraries to enhance service delivery, streamline transactions, and provide faster access to information resources.

However, despite this shift toward digitization, numerous public libraries continue to face serious challenges related to cybersecurity and record accuracy. Many existing systems, especially those that are outdated or poorly maintained, lack proper security mechanisms, making them vulnerable to data breaches, unauthorized access, and data manipulation (Lewis & Lipkowitz, 2025). Sensitive information, such as borrower records, transaction history, and inventory details, can be exposed when security controls are weak or improperly implemented (Raghavaiah & Karnati, 2025).

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In addition, manual data entry and the absence of validation mechanisms frequently lead to inaccurate records, duplication of data, and inconsistencies within the system. These issues not only compromise the integrity of the library's information but also disrupt library operations, resulting in misplaced books, incorrect borrower information, and unreliable reporting. Therefore, there is a growing need for a secure and reliable library management system that integrates cybersecurity measures and data validation techniques to ensure both data protection and record accuracy.

Given these challenges, there is a clear need for a Cybersecurity-Enhanced Library Management System that integrates secure authentication, role-based access control, encryption, audit trails, and data validation mechanisms to protect sensitive information and improve record accuracy (Akor, 2024; Purdue University, 2025). By combining cybersecurity principles with structured system development, such a system can serve as a practical application of knowledge in software engineering, database management, and information security.

1.2 Statement of the Problem

Despite the availability of modern library technologies, many libraries still rely on manual or partially automated systems in managing records and daily transactions. These outdated practices result in inefficiencies, inaccuracies, and security vulnerabilities.

This study aims to address the following problems:

1. Library records and transactions are still managed manually, leading to slow processing and operational inefficiencies.
2. There is difficulty in accurately tracking borrowed and returned books, resulting in misplaced or unmonitored inventory.
3. Manual recording increases the risk of data inaccuracies, duplication, and inconsistencies in library records.
4. Existing systems lack adequate cybersecurity measures, making them vulnerable to unauthorized access, data breaches, and data manipulation.
5. There is no reliable backup and recovery mechanism, increasing the risk of data loss in case of system failure or cyberattacks.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop a secure, efficient, and user-friendly web-based Library Management System that improves library operations by enhancing inventory tracking, ensuring accurate and reliable record management, and streamlining the borrowing and returning processes. The system aims to reduce manual errors and increase operational efficiency through automation and real time data monitoring.

Furthermore, the study seeks to integrate essential cybersecurity measures such as secure authentication, role-based access control (RBAC), data encryption, and audit logging to protect sensitive library information from unauthorized access, data breaches, and manipulation. The system will also incorporate data backup and validation mechanisms to ensure data integrity, availability, and reliability.

Overall, the proposed system aims to provide a practical and secure solution that supports librarians, students, and community users by improving accessibility, efficiency, and data security in public library management.

1.3.2 Specific Objectives

This study aims to:

1. To develop a secure and user-friendly web-based application interface that improves system accessibility and usability.
2. Implement cybersecurity measures to protect library records.
3. To design a data backup system to prevent loss of records.
4. Improve inventory tracking and monitoring of books in the library.
5. Provide real-time visibility of book availability for borrowing.

1.4 Scope and Delimitation

Scope:

This study focuses on the design, development, and evaluation of a web-based Cybersecurity-Enhanced Library Management System intended for public libraries in Iloilo and nearby communities in the Philippines. It aims to improve library operations by providing essential functionalities such as user authentication, role-based access control (RBAC) for librarians and borrowers, efficient book inventory management, and streamlined borrowing and return processes.

The system also integrates cybersecurity mechanisms including data encryption to protect sensitive information, audit trails for monitoring system activities, real-time book availability checking for accurate records, and automated data backup to ensure data integrity and system reliability.

Furthermore, the study evaluates the system based on usability, security effectiveness, and operational efficiency, as assessed by elementary, high school, and college users to determine its overall performance and practicality in real-world library settings.

Delimitation:

The primary scope of this research is centered on the design, development, and preliminary testing of a cybersecurity-enhanced management system tailored specifically for the operational needs of a single pilot public library in Iloilo. By confining the study to a single local institution, the researcher can ensure a more granular and focused evaluation of the software's effectiveness in a controlled environment before considering broader applications.

The technical boundaries of this study are strictly limited to software-based architectural solutions. The development focuses on internal logic, database integrity, and user-level security protocols to prevent record inaccuracies and data breaches. Consequently, the research explicitly excludes physical infrastructure and hardware-centric security measures.

While recognized as vital to holistic security, elements such as Closed-Circuit Television (CCTV) systems, biometric access hardware, and enterprise-grade physical network firewalls are outside the developmental parameters of this specific web application.

1.5 Significance of the Study

The proposed system will benefit the following:

- **Students** (elementary, high school, college)

The system allows students to easily check the availability of books in real time, reducing the time and effort required to search for resources and improving their overall learning

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experience. .

- **Local residents of Iloilo and nearby communities**

Community members will benefit from improved access to library information, enabling them to check available materials without physically visiting the library.

- **Librarians**

The system helps librarians manage records more efficiently by automating inventory tracking and borrowing processes. It reduces manual workload, minimizes errors, and enhances service delivery through a more organized and secure system.

- **Future Researchers**

This study may serve as a reference for future researchers interested in cybersecurity, web-based systems, and library management system development. It provides insights into integrating security features such as RBAC, encryption, and audit trails into practical applications.

1.6 Definition of Terms

Library Management System - A Library Management System (LMS), also known as an Integrated Library System (ILS), is an enterprise resource planning software designed to automate the lifecycle of library materials. By migrating from manual ledgers to a centralized digital database, institutions can drastically improve operational efficiency, data accuracy, and user engagement.

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Agile Methodology - The Agile methodology is an approach that divides work into phases, emphasizing continuous delivery and improvement. Agile benefits teams by enabling adaptive planning, rapid execution, and ongoing evaluation, leading to more responsive and successful outcomes.

Audit Trails -. It shows a sequential record of all the activity on a specific system. By reviewing audit logs and correlated audit trails, systems administrators can track user activity, and security teams can investigate breaches and ensure compliance with regulatory requirements

Authentication - Authentication is the process of validating the identity of a user by usually obtaining some sort of credentials and making sure that they match the user's provided login information, in order to produce some type of output which shows some sort of validation that the user is who they say they are.

Borrower Management - The process of handling, updating, and monitoring borrower information, membership status, and borrowing activities within the library system.

Borrower Record - An official record containing key information about each borrower, such as their personal details, membership identification, and the items they have checked out.

Brute Force Protection - a cybersecurity mechanism that defends accounts and systems against automated, trial-and-error attempts to guess passwords, encryption keys, or login credentials

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CAPTCHA - Stands for “Completely Automated Public Turing”, tools you can use to differentiate between real users and automated users, such as bot.

Cyber Attack - A cyber attack is a set of actions performed by threat actors, who try to gain unauthorized access, steal data or cause damage to various computing systems.

Cybersecurity - A practice of protecting systems, networks, and data from cyber threats such as unauthorized access and data breaches

Database - A structured collection of data where book records, user information, and library transactions are stored.

Data Backup – A security feature that saves copies of library records to prevent data loss.

Data Breach - Incident where sensitive information is accessed without authorization

Data Integrity - A concept and process that ensures the accuracy, completeness, consistency, and validity of an organization’s data.

Data Manipulation - Alteration or tampering of data, that affects accuracy and reliability

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Data Validation - Ensuring that the data entered is accurate and in correct format.

Database Management - A process of implementing and maintaining databases to store and retrieve information efficiently.

Disaster Recovery Plan - Is a documented, structured approach that describes how an organization can quickly resume operations after an unplanned incident

Encryption - Changes data into a secure format to prevent unauthorized access.

Encryption Key - a variable value that is applied using an algorithm to a string or block of unencrypted text to produce encrypted text or to decrypt encrypted text.

Flask - Flask is a lightweight WSGI web application framework. It is designed to make getting started quick and easy, with the ability to scale up to complex applications.

HTTPS - Hypertext transfer protocol secure (HTTPS) is the secure version of HTTP, which is the primary protocol used to send data between a web browser and a website.

Information Security - Protects information from unauthorized access, disclosure, disruption, modification, or destruction.

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Mitigation - A decision, action, or practice intended to reduce the level of risk associated with one or more threat events, threat scenarios, or vulnerabilities

Multi-Factor Authentication - Multi-factor Authentication (MFA) is an authentication method that requires the user to provide two or more verification factors to gain access to a resource such as an application, online account, or a VPN .

OTP - A one-time password (OTP) is an automatically generated numeric or alphanumeric string of characters that authenticates a user for a single transaction or login session.

Password Hashing - A one way function that transforms a plaintext password into a fixed-length string of characters using a hashing algorithm.

Phishing - A cyber attack that refers to an attempt to steal sensitive information, typically in the form of usernames, passwords, credit card numbers, bank account information, or other important data in order to utilize or sell the stolen information.

Ransomware - Ransomware is a type of malicious software or malware that prevents you from accessing your computer files, systems, or networks and demands you pay a ransom for their return.

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Real-Time Availability - System ability to instantly display the current status of books, either available or borrowed.

Role-Based Access Control (RBAC) - A security control that limits access based on roles

Session Tracking - Session Tracking is the process of monitoring and recording individual user visits to a website or application.

Sql Injection - a code injection technique used to modify or retrieve data from SQL databases.

Software Engineering - The practice of designing, developing, testing and maintaining software applications properly.

Transaction History - Records of borrowing and returning activities that library uses do.

User authentication refers to the process of verifying the identity of a user attempting to gain access to a computer network, system or device

User Authentication - User authentication refers to the process of verifying the identity of a user attempting to gain access to a computer network, system or device

Unauthorized Access - An entry in a system or data without permission that leads to data breaches or manipulation.

Validation Mechanism - A systematic process that checks whether the data entered into an application or system meets the specified requirements.

Vulnerability - Weakness in an information system, system security procedures, internal controls, or implementation that could be exploited or triggered by a threat source .

Web-Based - Refers to a system that runs on a web browser, accessible via the internet.

XAMPP - A free, open-source, cross-platform software stack for local web development. It bundles the pieces you need to run a basic website or web app on your own machine: a web server, a database server, and scripting support.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Breeding (2024) - Ransomware Attacks on Global Library Systems

Breeding (2024) provides an in-depth examination of ransomware attacks that have severely disrupted major library operations worldwide, serving as a critical wake-up call for libraries of all sizes to strengthen their cybersecurity measures. The British Library (BL), recognized as one of the world's most prestigious and well-resourced institutions, experienced what Breeding

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describes as a near "worst case scenario" when attackers compromised its systems on October 28, 2023. The assault affected virtually every component of the BL's technology infrastructure, including the Integrated Library System (ILS), online catalog, platforms supporting extensive digital collections, request and retrieval services, and even the Wi-Fi network. As a result, library users at both the main St. Pancras building in London and the remote Boston Spa facility lost access to physical collections entirely, rendering the world's largest library unable to perform its core functions for months.

Recovery proved extraordinarily challenging, with the online catalog only partially relaunched on January 15, 2024 without the ability to request materials and full restoration requiring extensive additional work. Breeding notes that the operational disruptions were compounded by the public release of sensitive internal data, including personal and financial details of library staff and users, prompting the BL to contact affected individuals with mitigation advice. This ransomware attack, claimed by the Rhysida cybercriminal group, exemplifies how such threats exploit infrastructure vulnerabilities to encrypt data, making it inaccessible without the attackers' decryption keys, which they demand in bitcoin through anonymous wallets. Beyond ransom, perpetrators threatened to auction stolen data or release it publicly, amplifying the damage.

Breeding extends the analysis to other high-profile cases, including the Toronto Public Library (TPL) attack in October 2023, where the website, catalog, and services remained unavailable into mid-January 2024, limiting users to external vendor content only. The 2022 Baker & Taylor ransomware incident, though resolved in 17 days, further illustrates the supply chain risks facing libraries. The author details recovery complexities: while experts advise against paying ransoms

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to avoid encouraging future attacks, 56% of surveyed educational institutions did so, facing additional costs for system audits to detect backdoors or malware. Non-payment strategies demand robust disaster recovery, including fresh OS/software reinstalls and data restoration from uncompromised backups challenges Breeding warns can cascade if backups themselves become encrypted.

On attack methodologies, Breeding explains Rhysida's use of software toolkits and social engineering, where intruders gain initial access via unpatched vulnerabilities or phishing, then linger undetected to map systems before encryption deployment. The rise of "ransomware as a service" syndication lowers barriers for less skilled criminals, broadening the threat landscape. For prevention, Breeding stresses proactive measures: continuous OS/application updates (countering libraries' tendency to delay patches for stability), dedicated security personnel for local servers, regular audits, frequent isolated backups on physically separate servers with strict access controls, and preference for vendor-hosted systems offering superior monitoring. Critically, personnel training against social engineering phishing via email, texts, or WhatsApp addresses the human weak link, as institutional filters may miss personal accounts.

These findings underscore that public-serving organizations like libraries face relentless targeting, with large institutions seen as ransom-rich and smaller ones vulnerable despite lower profiles. Breeding concludes that technology's benefits come with vigilance costs, urging libraries to close weaknesses before breaches like the BL's occur, directly validating the need for integrated safeguards in modern library management systems.

2.2 Local Studies

The study titled *Library Systems Development: Utilization, Usability and Impact for Private Schools in Negros Occidental, Philippines* examined the implementation and effectiveness of a Library Management System (LMS) in selected private schools in Negros Occidental. The researchers focused on evaluating the utilization, usability, and impact of the system on library operations and services. The study revealed that the use of a Library Management System significantly improved the efficiency and organization of library processes. Traditional manual methods of recording and monitoring books were replaced with computerized processes, allowing librarians to manage resources more accurately and efficiently. Daily library tasks such as cataloging, borrowing, returning, and tracking books became faster and more convenient through the use of the system.

The study also highlighted that the LMS improved accessibility of information for students, teachers, and librarians. Users were able to search for books and other learning resources more quickly, reducing the time spent on manual searching and paperwork. The automated system minimized human errors in recording transactions and maintaining inventory records. In addition, the system enhanced the monitoring of book availability, overdue materials, and user transactions, which contributed to better management of library resources. The researchers noted

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that the implementation of the LMS increased productivity and improved the overall quality of library services in schools.

Despite the positive outcomes, the study also identified several challenges encountered during the implementation of the Library Management System. One of the major issues was the limited technical knowledge and system familiarity among users. Some librarians and students experienced difficulty navigating the system because they were not fully trained in using digital platforms. Insufficient training and lack of technical support affected the usability of the system, especially during the early stages of implementation. The study also mentioned resistance to change as another challenge, since some users were more comfortable with traditional manual processes and were hesitant to adapt to new technologies. Technical problems such as system maintenance, internet connectivity issues, and hardware limitations were also observed. Nevertheless, the study concluded that the advantages of using a Library Management System outweigh the challenges because it greatly improves operational efficiency, accessibility, and service delivery in libraries.

In addition, the study titled *Navigating Cybersecurity Challenges in the Era of Digital Transformation: Threats and Mitigation Strategies in the Philippines* explored the growing cybersecurity risks faced by organizations as they transition from manual systems to digital platforms. The researchers explained that digital transformation provides many benefits, including faster transactions, easier access to information, and improved operational processes. However, the increasing reliance on digital systems also exposes organizations to various cyber threats and security vulnerabilities. The study emphasized that organizations in the Philippines

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are becoming more vulnerable to cyberattacks such as data breaches, phishing attacks, malware infections, identity theft, and unauthorized access to sensitive information.

The study further discussed that many organizations still lack proper cybersecurity awareness, security policies, and effective protection mechanisms. Weak passwords, poor authentication methods, lack of encryption, and insufficient monitoring systems make digital platforms more susceptible to attacks. According to the researchers, cyber incidents can lead to data loss, financial damage, operational disruption, and loss of user trust. To address these risks, the study recommended the implementation of strong cybersecurity measures such as secure authentication systems, multi-factor authentication, data encryption, regular system monitoring, firewall protection, and cybersecurity awareness training for users and administrators. The study also highlighted the importance of adopting proactive cybersecurity strategies rather than relying only on reactive responses after incidents occur.

These local studies demonstrate that Library Management Systems greatly improve the efficiency, usability, and accessibility of library operations. However, they also reveal that challenges related to system adoption, technical limitations, and cybersecurity threats remain significant concerns. The findings of these studies support the need for a secure and efficient web-based Library Management System that not only improves library services and record management but also integrates strong cybersecurity measures to protect sensitive data and ensure system reliability.

2.3 Related Literature

Related literature provides the conceptual and theoretical foundation of the study by presenting findings from various articles, journals, books, and credible online sources. These studies are essential in understanding the role of information systems, software development practices, and technologies used in the development of a secure web-based Library Management System.

Information systems play a vital role in the automation and improvement of organizational processes by collecting, processing, storing, and distributing data efficiently. According to Bhati et al. (2024), web based Library Management Systems significantly improve library operations by automating processes such as cataloging, borrowing, returning, and inventory tracking. Their study emphasizes that information systems reduce manual workload, minimize human errors, and improve the speed and accuracy of data retrieval in academic libraries. This shows that LMS platforms are essential tools in modernizing library services and improving user satisfaction.

Furthermore, cloud-based and digital information systems have become increasingly important in improving accessibility and resource sharing. Hazarika and Konch (2025) explain that modern digital library systems enhance availability and efficiency of information services but also require strong security mechanisms due to increased exposure to cyber threats. This highlights that while information systems improve operations, they must also be designed with proper security integration.

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Software development is the process of designing, building, testing, and maintaining software systems to ensure quality, functionality, and reliability. Research in software engineering emphasizes the importance of structured development approaches to ensure system stability and maintainability. Modern studies show that following systematic development practices helps reduce system errors and improves long-term performance, especially in web-based applications.

In addition, cybersecurity must be considered during the software development process. Systems that handle sensitive data, such as Library Management Systems, must be designed with secure coding practices, proper architecture, and vulnerability prevention techniques. Nandagaoli and Lihitakar (2019) highlight that poorly designed web-based systems are highly vulnerable to attacks such as unauthorized access, data breaches, and system exploitation. This reinforces the importance of integrating security into the software development lifecycle rather than treating it as an afterthought.

Modern Library Management Systems are commonly developed using web-based technologies that allow accessibility, scalability, and user-friendly interfaces. Web development technologies such as HTML, CSS, and JavaScript are used for designing interactive and responsive user interfaces, while backend frameworks such as Python Flask are used to handle system logic and database communication.

Bhati et al. (2024) explain that web-based LMS platforms improve operational efficiency by integrating database systems that manage large volumes of library data. MySQL is commonly used as a relational database management system due to its reliability, speed, and ability to handle structured data such as book records, user accounts, and transaction logs.

Security technologies are also essential in modern systems. Encryption, authentication, and access control mechanisms are widely used to protect sensitive information. Hazarika and Konch (2025) emphasize the importance of implementing security measures such as encryption and multi-factor authentication in digital systems to prevent unauthorized access and data breaches. These technologies ensure that only authorized users can access and modify system data.

2.4 Synthesis of the Reviewed Literature

The reviewed literature provides a comprehensive understanding of the role of Library Management Systems (LMS), software development practices, and cybersecurity integration in modern digital environments. Across foreign, local, and related studies, there is a consistent emphasis on the importance of automation, system efficiency, and data protection in library operations. However, the literature also highlights significant challenges, particularly in terms of cybersecurity vulnerabilities, system implementation difficulties, and limited technical capacity in educational institutions.

Foreign studies strongly emphasize the critical risks associated with the digital transformation of library systems. Breeding (2024) documented the severe consequences of ransomware attacks on major institutions such as the British Library and Toronto Public Library. These incidents resulted in prolonged system downtime, restricted access to catalogs and digital collections, and exposure of sensitive user and institutional data. The studies further explain that cybercriminals

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exploit system vulnerabilities through phishing, unpatched software, and social engineering techniques. Once access is gained, attackers encrypt or steal data and demand ransom payments, often causing long-term operational disruption. These findings highlight that even well-funded and technologically advanced libraries are not immune to cyber threats, emphasizing the urgent need for robust cybersecurity frameworks in library systems of all sizes.

Local studies present a similar yet context-specific perspective on Library Management Systems within the Philippines. Research conducted in private schools in Negros Occidental shows that LMS implementation significantly improves library operations by automating cataloging, borrowing, returning, and inventory management processes. These systems enhance efficiency, reduce manual workload, and improve accuracy in record keeping. Users are able to access information more quickly, which improves the overall delivery of library services. However, despite these benefits, local studies also reveal persistent challenges such as insufficient technical knowledge among users, lack of proper training, resistance to technological change, and occasional system maintenance and connectivity issues. These limitations often reduce the full effectiveness of LMS adoption in educational institutions.

In addition, cybersecurity-related local studies highlight that many organizations in the Philippines remain highly vulnerable to cyber threats due to weak security infrastructure and limited awareness of digital risks. Common issues include poor password practices, lack of encryption, inadequate access control systems, and insufficient monitoring mechanisms. These weaknesses expose systems to risks such as unauthorized access, phishing attacks, malware infections, and data breaches. The studies further emphasize that cybersecurity incidents can lead

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to severe consequences, including loss of sensitive data, disruption of operations, financial damage, and reduced trust from users. As a result, there is a strong recommendation for organizations to adopt proactive cybersecurity strategies, including multi-factor authentication, secure system design, regular monitoring, and user awareness training.

Related literature further strengthens these findings by explaining the theoretical and technical foundations of information systems and software development. Information systems literature explains that systems are designed to collect, process, store, and distribute information efficiently to support organizational operations and decision-making. In the context of libraries, LMS platforms serve as essential tools for managing large volumes of data and improving accessibility for users. Studies such as Bhati et al. (2024) confirm that web based LMS solutions enhance operational efficiency by reducing manual processes and improving data accuracy and retrieval speed.

Software development literature also emphasizes the importance of structured and systematic approaches in building reliable applications. Proper software engineering practices ensure that systems undergo careful planning, design, implementation, testing, and maintenance to achieve high quality and long-term sustainability. Furthermore, modern development practices highlight that security must be integrated throughout the software development lifecycle rather than added as an afterthought. This is particularly important for systems that handle sensitive user and institutional data, such as library management systems.

Technological literature supports the use of modern web-based tools and frameworks in building efficient and scalable LMS platforms. Frontend technologies such as HTML, CSS, and

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JavaScript enable the creation of interactive and user-friendly interfaces, while backend frameworks such as Python Flask provide flexibility and efficient handling of system logic and database operations. MySQL is widely used as a relational database management system due to its reliability, speed, and ability to handle structured data such as book records, user accounts, and transaction logs. Additionally, security technologies such as encryption, authentication, and access control are essential in ensuring that only authorized users can access sensitive data and system functionalities.

Research Gap:

Despite the availability of advanced global LMS frameworks and numerous studies on digital library systems, there remains a significant gap in the development of secure, simplified, and cost-effective systems tailored for small-scale or resource-limited institutions in the Philippines. Foreign studies primarily focus on large and highly advanced libraries with strong technical infrastructure, while local studies mainly identify operational challenges and security weaknesses without providing fully integrated solutions. Furthermore, there is limited research that combines core library management functions with practical cybersecurity implementations in a single, lightweight web-based system suitable for academic environments.

How the Current Study Addresses the Gap:

Our study addresses the identified gaps by developing a secure web based Library Management System that integrates essential library functions with modern cybersecurity mechanisms. The proposed system provides features such as real-time inventory tracking, borrowing and returning

management, and user role management, while also implementing strong security measures including encryption, role-based access control, audit logging, and secure authentication processes.

Unlike complex enterprise-level systems that require high costs and advanced infrastructure, the proposed system is designed to be simple, efficient, and suitable for small academic and public libraries. It aims to bridge the gap between functionality and security by providing a practical solution that improves operational efficiency while protecting sensitive data from unauthorized access and cyber threats. Through this approach, the system ensures both usability and security, making it a suitable and sustainable solution for modern library environments in the Philippines.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study will use a developmental research design, which is a systematic and iterative approach focused on the design, development, implementation, and evaluation of the proposed web-based application. This research design is appropriate for studies that involve the creation of a new

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system or technology, particularly when the goal is to produce a functional and practical output such as a Library Management System.

Through this approach, the researchers will follow a structured process that includes identifying system requirements, designing the system architecture, developing the application, and continuously testing and refining it to ensure that it meets the intended objectives. The developmental research design also emphasizes continuous improvement, meaning that feedback gathered during testing and evaluation will be used to enhance the system's performance and usability.

This approach ensures that the proposed system is not only developed based on theoretical concepts but is also validated through actual implementation and evaluation. The system will be assessed in terms of its functionality, usability, reliability, efficiency, and overall performance, ensuring that it effectively supports library operations such as cataloging, borrowing, returning, and record management.

In addition, the developmental research design allows the researchers to integrate important features such as security mechanisms, user accessibility, and database management throughout the development process. This is particularly important for a Library Management System, as it involves handling sensitive user information and requires accurate and secure data processing.

3.2 System Development Methodology

The System will be developed using Agile Development Methodology ,a flexible and iterative approach that allows continuous improvement throughout the development process. Agile focuses on collaboration, adaptability, and delivering functional components in incremental stages. This methodology is suitable for the project as it enables frequent feedback, quick adjustments, and efficient problem-solving.

The Agile Development Model consists of the following phases:



1. Plan - In this phase, the project requirements are identified and analyzed. The development team collaborates with stakeholders to define system goals, features, and scope. A project plan is created, including timelines, tasks, and resource allocation.

2. Design - The system architecture and interface are designed based on the identified requirements. This includes creating wireframes, database structures, and system flow diagrams to visualize how the system will function.

3. Develop - During this phase, the actual coding and implementation of the system take place. Developers build the features based on the design specifications while ensuring code quality and functionality.

4. Test - The system is tested to identify and fix errors or bugs. Various testing methods such as unit testing, integration testing, and user testing are conducted to ensure the system meets the required standards and performs correctly.

5. Deploy - After successful testing, the system is deployed and made available to users. This phase includes installation, configuration, and initial user support.

6. Review - In the final phase, feedback is gathered from users and stakeholders. The development team evaluates the system's performance and identifies areas for improvement. These insights are used for the next iteration, making Agile a continuous cycle.

3.3 System Architecture

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The system operates as a secure 3-role web application:

- Members/Users (view books availability).
- Librarians (manage records).
- Admins (full system control).

All interactions flow through Python Flask backend enforcing cybersecurity controls before accessing MySQL database. Data encrypted via HTTPS with role-based access control (RBAC).

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Client Server Set up

Client Browser (HTML/CSS/JS) Users	Web Server (Python + Apache) (Login Security)	Xampp MySQL Database (Data)
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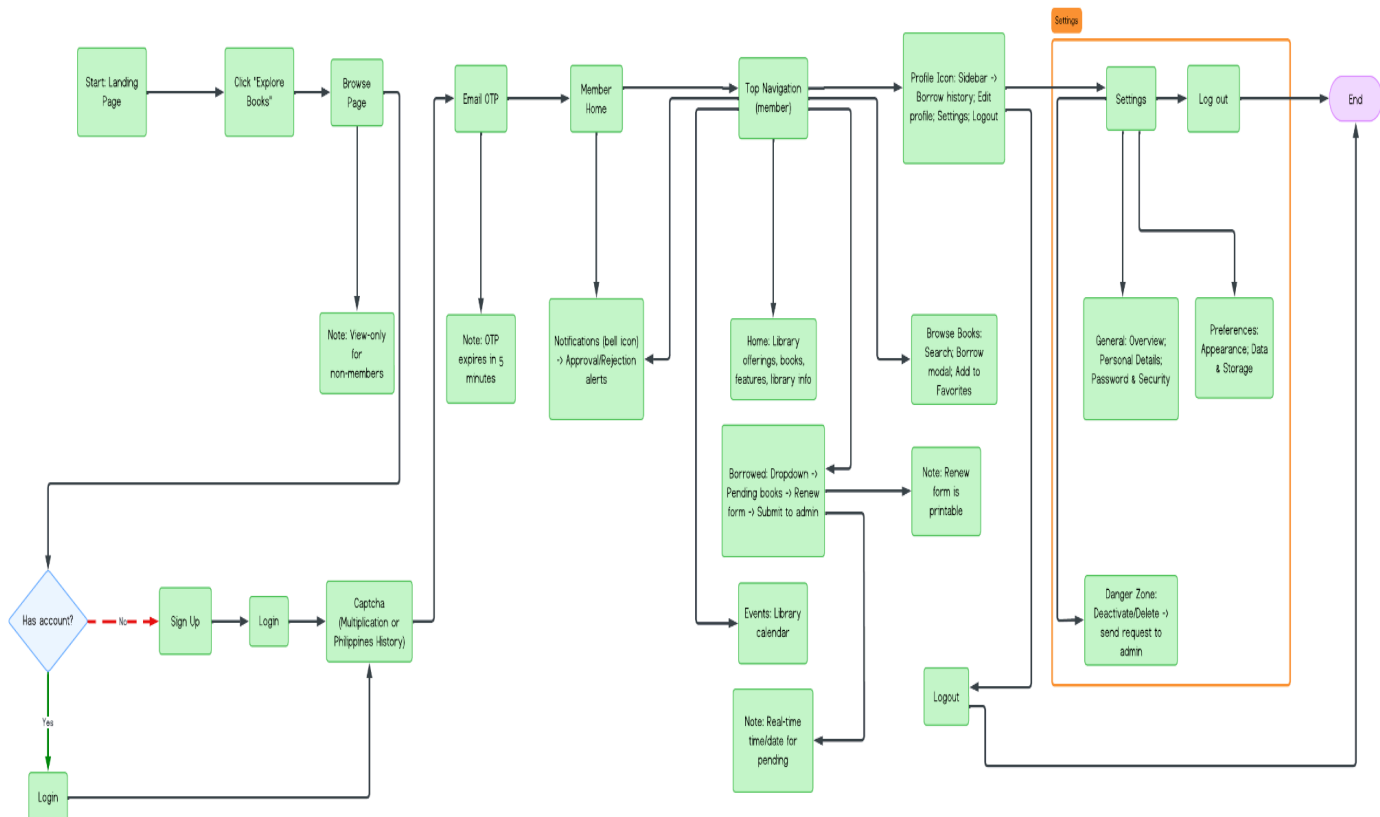
Client: Any web browser accessing the system

Server: localhost serves static files (HTML/CSS/JS)

Database: Stores books, borrowers, transactions, audit logs

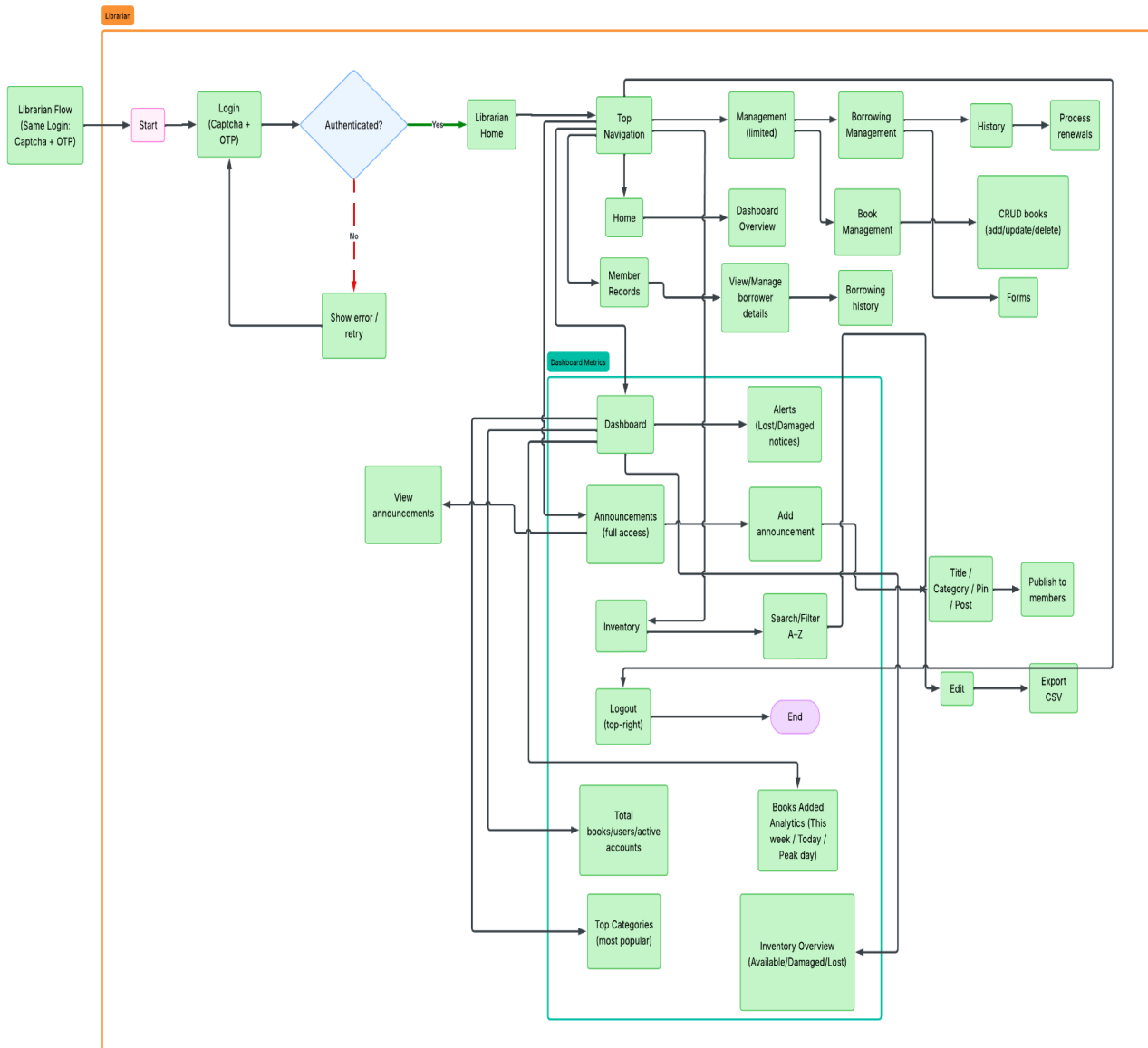
User Interaction Flow

Member Flow:



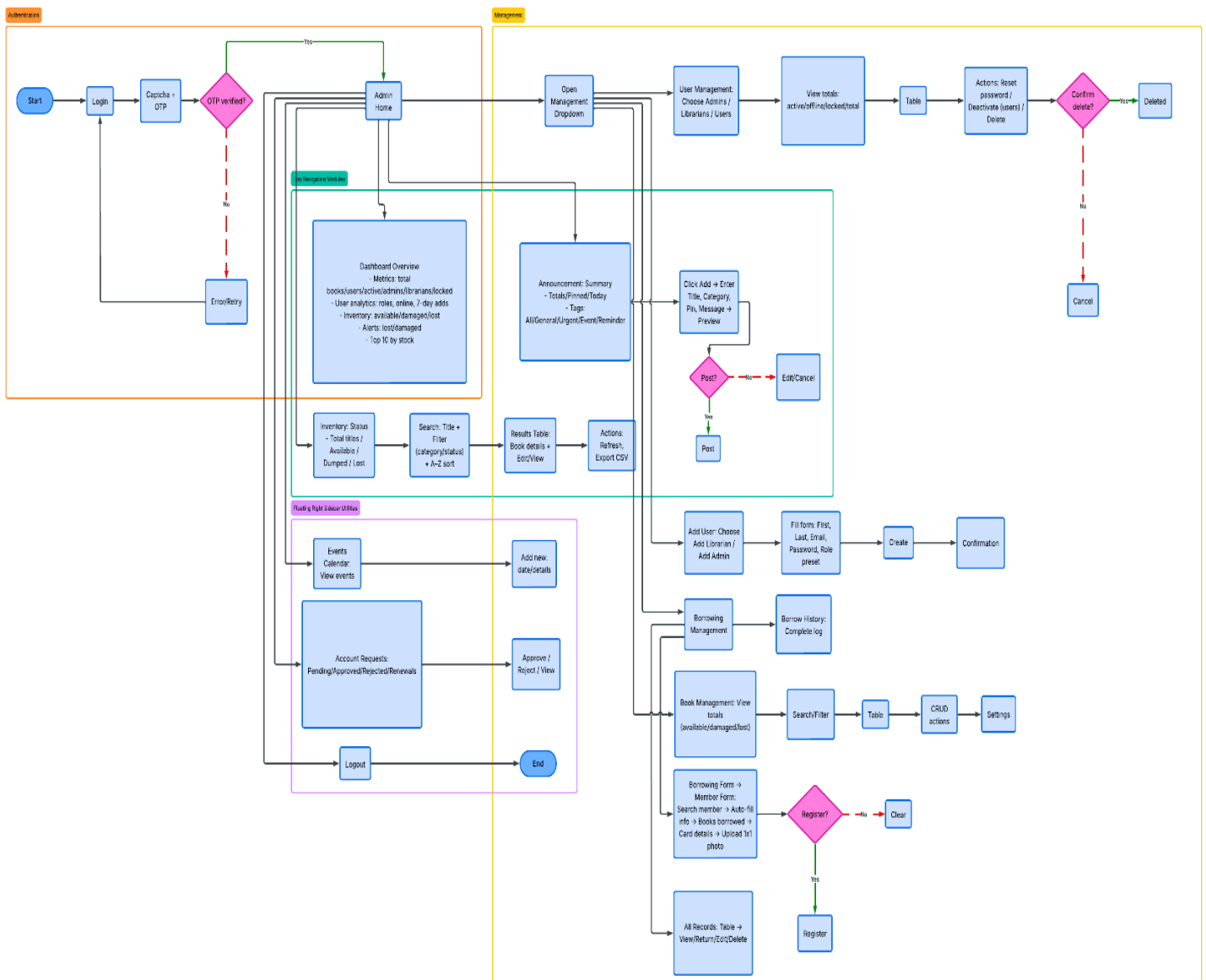
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Librarian Flow:



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Admin Flow:



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Security Implementation

Control	Implementation	Purpose
HTTPS	SSL/TLS certificate	Encrypt traffic
RBAC	Flask session roles	Access restriction
Encryption	Fernet AES-128-CBC	Data at rest
OTP / MFA	Email-based 6-digit OTP	Multi-factor authentication
Brute Force Protection	Account lockout after 5 failed attempts (15 min)	Prevent credential stuffing
CAPTCHA	Math & trivia challenge on login	Bot prevention
Password Policy	Bcrypt hashing + complexity regex	Secure credential storage
Password Reset	Signed time-limited token (1 hr)	Secure recovery flow
Input Sanitization	HTML tag stripping + length limits	XSS / overflow prevention
Approval Workflow	Admin approval before account activation	Unauthorized access prevention
File Upload Validation	Extension whitelist + size limits	Malicious file prevention
Session Tracking	last_seen timestamp + auto-offline	Session awareness
Audit / Error Handling	DB rollback on exception + error handlers	Data integrity & accountability
Backup	Scheduled executable backup	Data recovery

3.4 Tools and technologies Used

- Frontend: HTML, CSS, JavaScript/Bootstrap
- Backend: Python /Flask
- Database: Xampp/MySQL
- Development Tools: Visual Studio Code

3.5 Data Gathering Techniques

- Interview with librarians to understand book management and borrowing processes works.
- Checked existing records like book lists and borrowing logs to understand how information is stored and managed.
- Observed how books are manually managed in the library, including how they are recorded and borrowed.

3.6 Testing Procedure

- **System Testing** – Tested the whole system to make sure it works properly.

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- **Acceptance Testing** - Have the actual librarian or admin from the library validate that the system meets their real-world operational needs before deployment.
- **User Testing** – Let users try the system to see if it is easy to use and understand.
- **Usability Testing** Instead of simply allowing users to try the system, the study will involve real target users such as students and librarians performing specific tasks in the system. After completing these tasks, they will evaluate how easy the system is to use by answering a usability questionnaire. This helps measure how user-friendly the system is and supports the evaluation of its usability.

3.7 Evaluation Criteria

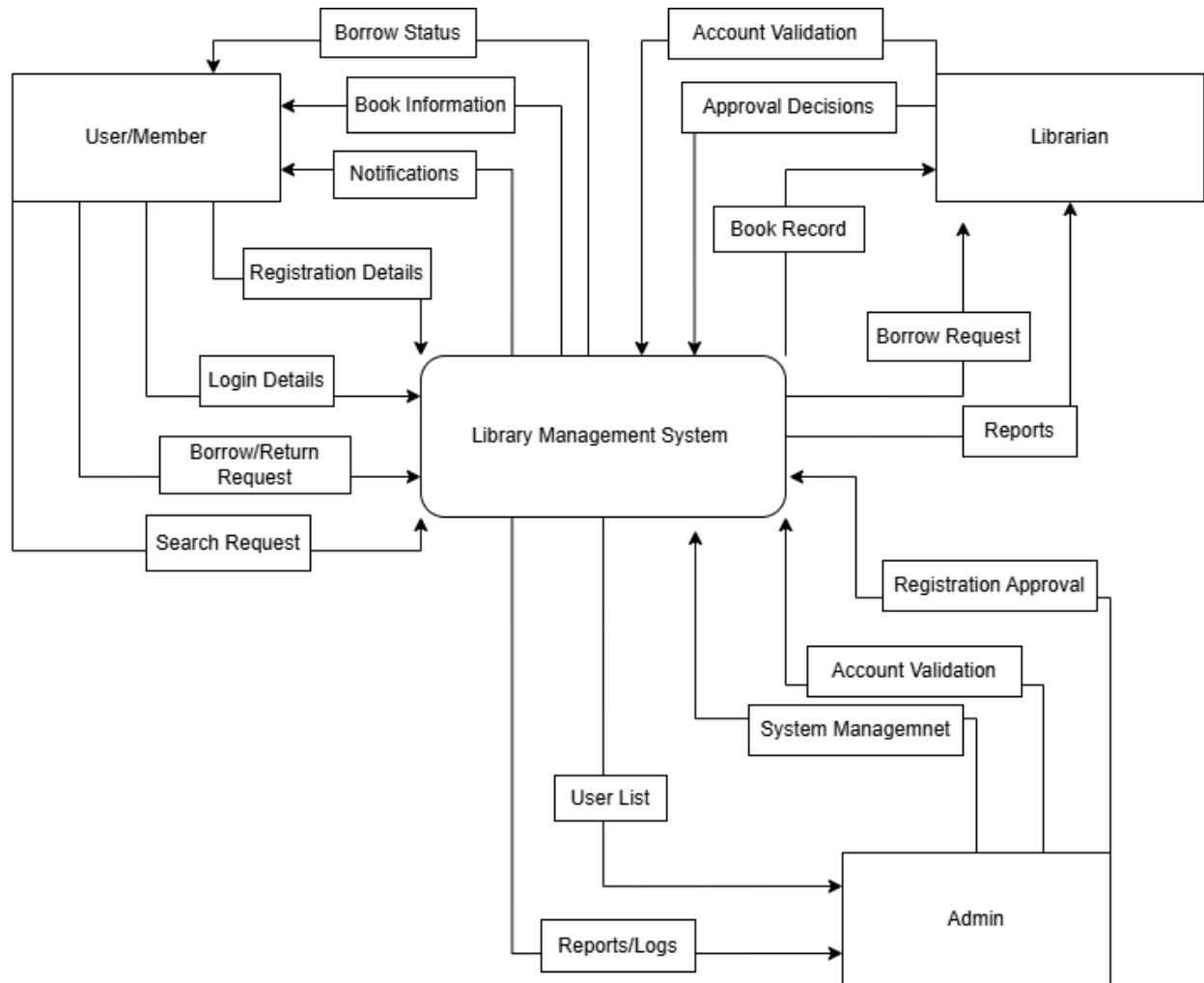
The system will be evaluated based on the following criteria:

- Accuracy of books status
- Efficiency
- Usability
- Reliability
- Functionality
- Availability

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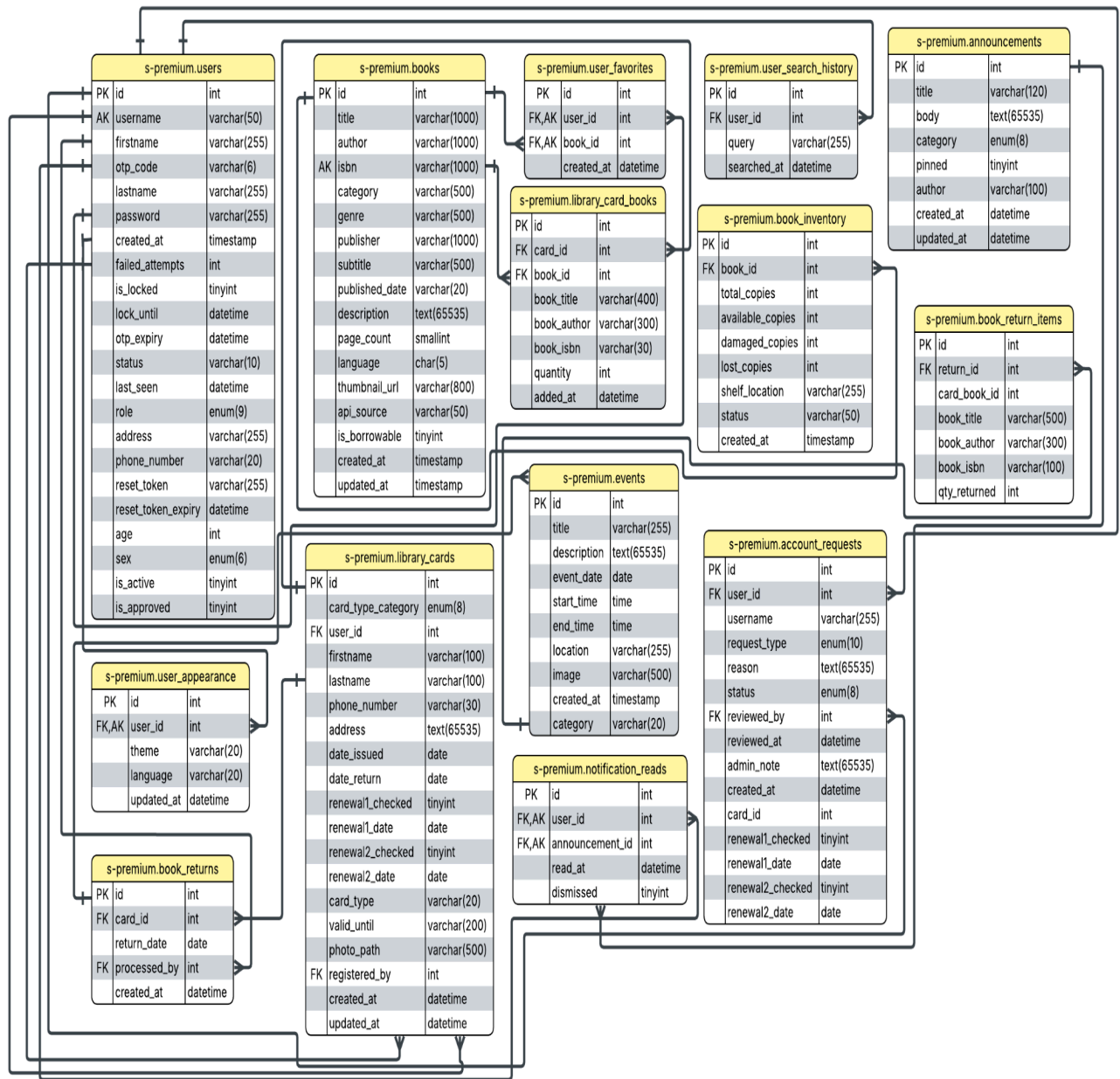
Data Flow Diagram (DFD – Level 0)

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Entity Relationship Diagram (ERD)



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Entities:

Users (id, username, firstname, lastname, otp_code, password, created_at, failed_attempts, is_locked, lock_until, otp_expiry, status, last_seen, role, address, phone_number, reset_token, reset_token_expiry, age, sex, is_active, is_approved)

user_appearance (id, user_id, theme, language, updated_at)

user_favorites (id, user_id, book_id, created_at)

user_search_history (id, user_id, query, searched_at)

books (id, title, author, isbn, category, genre, publisher, subtitle, published_date, description, page_count, language, thumbnail_url, api_source, is_borrowable, created_at, updated_at)

book_inventory (id, book_id, total_copies, available_copies, damaged_copies, lost_copies, shelf_location, status, created_at)

book_returns (id, card_id, return_date, processed_by, created_at)

book_return_items (id, return_id, card_book_id, book_title, book_author, book_isbn, qty_returned)

events (id, title, description, event_date, start_time, end_time, location, image, created_at, category)

library_cards (id, card_type_category, user_id, firstname, lastname, phone_number, address, date_issued, date_return, renewal1_checked, renewal1_date, renewal2_checked, renewal2_date, card_type, valid_until, photo_path, registered_by, created_at, updated_at)

library_card_books (id, card_id, book_id, book_title, book_author, book_isbn, quantity, added_at)

notification_reads (id, user_id, announcement_id, read_at, dismissed)

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account_requests (id, user_id, username, request_type, reason, status, reviewed_by, admin_note, created_at, card_id, renewal1_checked, renewal1_date, renewal2_checked, renewal2_date)

announcements (id, title, body, category, author, pinned, created_at, updated_at)

Relationships:

- A User can have a Library Card
- A User can make an Account Request.
- An Admin checks Account Requests.
- An Admin can register Library Cards.
- Librarians can handle Book Returns.
- A User can read Announcements.
- A User can favorite Books.
- A User can have Search History.
- A User has Appearance Settings.
- A Library Card can borrow Books.
- A Book has an Inventory.
- A Book has one Inventory record.

Use Case Diagram

Actors:

- Users
- Borrowers
- Admin
- Librarians

Use Cases:

Users / Member

- Explore Books
- Sign Up
- Login
- Browse Books
- Search Books
- Borrow Book
- Add to Favorites
- View Borrowed Books
- Renew Book
- View Notifications
- View Events
- Manage Profile

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- Edit Profile
- Change Password
- Adjust Settings
- Request Account Deletion
- Logout

Librarian

- Login
- View Dashboard
- View Member RecordsBorrowing Mana
- Manage Member Records
- Borrowing Management
- Process Renewals
- View Borrow History
- Book Management (Add/Edit/Delete)
- Manage Inventory
- View Alerts
- Manage Announcements
- Search/Filter Books
- Export Data (CSV)
- Logout

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Admin

- Login
- View Dashboard
- View Analytics
- Manage Users (Admin/Librarian/Member)
- Add User (Admin/Librarian)
- Reset Passwords
- Deactivate/Delete Accounts
- Borrowing Management
- Process Borrow Forms
- View Borrow History
- Book Management (CRUD)
- Manage Inventory
- Manage Announcements
- View Reports
- Handle Account Requests
- Approve/Reject Requests
- Manage Events Calendar
- Export Data

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